

Harini Vinu

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EDUCATION

Master of Science, Computer Science

New York University, New York

September 2024 — May 2026

GPA: 3.74/4.00

Courses: Data Structures & Algorithms, Distributed Systems, Big Data, Cloud Computing, Blockchain, Machine Learning, Deep Learning

Bachelor of Technology, Computer Science

SRM Institute of Science and Technology

September 2020 — May 2024

GPA: 3.8/4.00

Courses: Artificial Intelligence, OOPs, Operating Systems, Software Engineering, Computer Network, System Design, Database Systems

EXPERIENCE

Software Engineering Intern

Qualitest – CoCo (AI-driven Test Optimization Platform)

May 2025 — August 2025

New York, United States

- Developed backend services for AI-driven test orchestration using **Python**, **Redis** (caching, queue-based processing), and **MongoDB**.
- Enabled reliable distributed job scheduling across **100+** concurrent tasks across large-scale automated workflows.
- Engineered scalable data processing pipelines for large test datasets using asynchronous queue-based processing and optimized aggregation logic, reducing execution time by **40%** in high-throughput distributed systems.
- Integrated **monitoring & fault-tolerant** retry mechanisms for distributed jobs, improving system reliability & reducing job failures by **30%**.
- Architected backend services to manage test workflows, improving automation scalability and service orchestration.

Software Engineering Intern

LOCOMeX

June 2025 — August 2025

New York, United States

- Designed data validation and processing services to handle **100K+** records, ensuring consistency across large-scale ingestion workflows.
- Optimized query execution and **REST APIs** performance by restructuring hierarchical data models and database access patterns.
- Built scalable microservices using event-driven architecture to propagate risk signals across supplier networks in real time.
- Implemented aggregation, transformation and normalization logic for large supplier datasets, improving ESG risk scoring accuracy and reducing latency by **30%**.

Big Data Engineer Intern

New York University

January 2025 — May 2025

New York, United States

- Engineered distributed data pipelines using **Apache Spark**, **Kafka**, & **Hadoop** to process **10M+** records across scalable clusters.
- Accelerated automation tools to streamline workflow reliability, cutting issue resolution time by **90%**.
- Optimized scalable data ingestion and processing modules with real-world datasets, improving distributed system efficiency by **25%**.
- Designed real-time streaming data ingestion systems using Kafka producers and consumers, enabling near-real-time analytics and reducing data latency for downstream reports.

Software Engineering Intern

Skill Vertex

December 2021 — February 2022

Chennai, India

- Built a hand detection and gesture recognition system with **Python**, **OpenCV**, and **Google AI tools**, improving detection accuracy by **40%**.
- Enhanced object tracking and optimized machine-learning pipelines using preprocessing and hyperparameter tuning, which cut false detections by **15%**, lowered latency by **25%**, and achieved real-time processing at 30 FPS on over 1,000 test samples.

PROJECTS

Dining Concierge Chatbot | AWS Lambda, API Gateway, DynamoDB GitHub ↗

September 2025 — October 2025

- Developed serverless microservices using **AWS (S3, API Gateway, Lambda, Lex, DynamoDB, Elasticsearch)** integrated with the **Yelp API** to deliver real-time restaurant recommendations through a conversational chatbot. Orchestrated asynchronous workflows using **SQS**, enabled automated personalized alerts via **SES**, and leveraged **CloudWatch** for monitoring, logging, and event-driven scheduling.

Real-Time Stock Market Analysis System | Kafka, Spark Streaming, Hadoop GitHub ↗

September 2024 — October 2024

- Designed a distributed streaming analytics platform using **Kafka**, **Spark Streaming**, & **Hadoop** to process live financial data feeds. Built backend services for computing technical indicators and storing high-volume time-series data in **PostgreSQL & InfluxDB**, enabling low-latency analytics. Deployed data ingestion pipelines & used **Grafana** for performance monitoring & analytics visualization.

Real-Time Video Monitoring and Alert System | AWS Kinesis, Lambda, DynamoDB GitHub ↗

January 2025 — May 2025

- Built a scalable real-time monitoring system using **AWS Kinesis**, **Lambda**, **SQS**, **DynamoDB**, and **Rekognition** to process live data streams and perform real-time image analysis, generating low-latency insights and alerts. Designed event-driven backend services to trigger notifications with millisecond-level latency and data storage layers for real-time monitoring dashboards. Implemented analytics pipelines using **Elasticsearch** and **AWS QuickSight** to analyze streaming event data and provide operational insights.

TECHNICAL SKILLS

Programming & Languages: Python, C++, Java, Scala, Go, JavaScript, Bash

Backend & APIs: Microservices Architecture, Spring Boot, FastAPI, Django, Flask, gRPC, GraphQL, REST APIs, OpenAPI, Swagger

Data & Machine Learning: TensorFlow, Scikit-learn, OpenCV, Computer Vision, ML Pipelines, Hadoop, Hive, Akka

Databases & Caching: MySQL, PostgreSQL, MongoDB, Redis, InfluxDB, SQLite, Hibernate

Cloud, DevOps, Distributed Systems & Monitoring: AWS (Kinesis, Step Functions, CloudFront, CloudFormation, Cognito, Rekognition), Serverless, Docker, Kubernetes, Terraform, Jenkins, Prometheus, Grafana, Splunk, Ansible, Linux/Unix

Certifications: NYU: CareerHub Data Science Bootcamp ; Accenture: AI; Infosys SpringBoard: Java Certified Foundation Associate